

Manasa Bobbili

Python Full Stack Developer

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Professional Summary:

- Full-Stack Python Developer with 5+ years of experience delivering scalable, cloud-native applications following SDLC methodology.
- Strong expertise in Python frameworks including Django, Flask, FastAPI, for building REST APIs, microservices, and backend systems.
- Skilled in frontend development using React.js, Angular, Vue.js, Material UI, HTML5, CSS3, and JavaScript (ES6+), delivering responsive and accessible UIs.
- Hands-on experience with SQL and NoSQL databases such as MySQL, PostgreSQL, MongoDB including data modeling, ETL pipelines, and analytics.
- Designed and implemented data processing and visualization workflows using Pandas, NumPy, Matplotlib, and HighCharts.
- Experienced in cloud platforms including AWS (EC2, S3, Lambda, RDS, ECS), Azure (App Service, Functions, Storage), and containerization with Docker and Kubernetes.
- Expertise in CI/CD pipelines and DevOps practices using Jenkins, GitLab CI, GitHub Actions, Terraform, for automated deployment and testing.
- Developed and integrated event-driven systems using Kafka, RabbitMQ, Celery, and ActiveMQ for real-time transaction processing and async data pipelines.
- Strong knowledge of web server deployment using Apache Tomcat, WebLogic, and Gunicorn, and API documentation with Swagger.
- Proficient in object-oriented programming (OOP), multi-threading, exception handling, and software design patterns.
- Skilled in Agile methodologies (Scrum, Kanban) with experience in JIRA, Confluence, sprint planning, and cross-functional collaboration.
- Adept at unit testing and test automation using Pytest, PyUnit, SonarQube, and Mockito, ensuring high-quality code and robust application performance.
- Strong problem-solving, analytical, and communication skills, capable of collaborating with developers, QA, product managers, and stakeholders.
- Experienced in mentoring junior developers, performing code reviews, and contributing to high-performing, quality-driven engineering teams.

Technical Skills:

Programming Languages: Python, JavaScript (ES6+), TypeScript.

Frameworks: Django, FastAPI, Flask

Web Development: Express.js, React.js, Angular, Vue.js, HTML5, CSS3, AJAX, Jquery, Bootstrap

Databases: MySQL, MongoDB, PostgreSQL, SQL Server, Cassandra

Cloud Platforms: AWS (EC2, S3, RDS, Lambda, EBS)

DevOps: Jenkins, Azure DevOps, Terraform, Docker, Kubernetes

Version Control: Git, GitHub, Bitbucket

Testing: PyUnit, Pytest, Selenium

API Design & Monitoring: Swagger, ELK Stack, CloudWatch

Project Management: JIRA, Confluence

Infrastructure & Servers: Nginx, Apache Tomcat, WebLogic Server, Gunicorn

Data Processing & Analysis: NumPy, Pandas

Methodologies: Agile, Scrum, TDD

Operating Systems: Linux, UNIX, macOS, Windows

Work Experience:

OneMain Financial, USA

October 2024 – Current

Python Full Stack Developer

- Designed and developed large-scale backend systems using Python, FastAPI, Django, and Flask, ensuring high availability, scalability, and fault tolerance.
- Built and maintained full-stack applications using React.js, JavaScript, HTML5, CSS3, and REST API integrations for seamless user experiences.
- Implemented authentication and authorization mechanisms using JWT, OAuth, or role-based access control.
- Architected document-processing and vector-search solutions using AWS OpenSearch, enabling low-latency retrieval across large datasets.
- Developed advanced data preprocessing pipelines to chunk structured/unstructured data and generate embeddings for vector databases.
- Built PySpark-based ETL pipelines to transform data from CSV, JSON, and Parquet into analytics-ready datasets.
- Developed RESTful Python APIs for event tracking, intelligent search, analytics, and real-time data processing.
- Processed complex datasets using NumPy, Pandas, Matplotlib, performing analysis, visualizations, and data-quality validation.
- Optimized database operations across MySQL, PostgreSQL, Oracle, Cassandra, using raw SQL, ORM frameworks, and indexing strategies.
- Developed responsive and interactive frontend interfaces using React.js, Angular, or Vue.js.
- Integrated frontend with backend APIs ensuring smooth data flow and user experience.
- Implemented dynamic content updates using AJAX, WebSockets, or server-sent events.
- Built automated testing suites using PyTest, Selenium, Jest, achieving high coverage across API and UI layers.
- Automated build, testing, and deployment pipelines using CI/CD tools like Jenkins, GitHub Actions, or GitLab CI.
- Deployed applications to AWS services like EC2, S3, RDS, Lambda, and Elastic Beanstalk for scalable cloud infrastructure.
- Worked on AWS data pipeline for Data Extraction, Transformation and Loading from the homogeneous or heterogeneous data sources.
- Built internal dashboards, analytics tools, and reporting views using Flask, Django, Plotly Dash, and various databases.
- Migrated legacy Python 2.x codebases to Python 3.11, improving performance, compatibility, and maintainability.
- Performed performance profiling and optimization of Python services running on AWS infrastructure.
- Created CI/CD pipelines using GitLab CI, GitHub Actions, Docker, Kubernetes, Terraform, and cloud-native deployment workflows.
- Enhanced observability using Prometheus, Grafana, and Sentry, improving monitoring, alerting, and incident response.
- Worked on multi-cloud environments, gaining experience with Azure and GCP services for flexibility and hybrid deployments.
- Collaborated with cross-functional teams in Agile/Scrum environments, participating in architecture reviews, code reviews, and mentoring junior engineers.

Environment: Python, FastAPI, Django, Flask, JavaScript, HTML5, CSS3, AWS (EC2, S3, RDS, Lambda, Elastic Beanstalk), PySpark, NumPy, Pandas, Matplotlib, MySQL, PostgreSQL, Oracle, Cassandra, React.js, Angular, Vue.js, AJAX, PyTest, Plotly Dash, Jenkins, Terraform, Docker, Kubernetes, Git.

Infosys, India
Software Engineer

March 2022 - December 2023

- Collaborated with business users during requirement-gathering workshops to analyze needs and document functional and technical specifications.
- Developed and maintained scalable web applications using Python, Django, Flask, and the MVC/MVT architectures.
- Wrote clean, modular, and object-oriented Python code, applying principles such as inheritance, encapsulation, and polymorphism.
- Built RESTful services and APIs using Django REST Framework, Flask, GraphQL, and Swagger documentation for seamless integration with internal and external systems.
- Utilized Django ORM, SQLAlchemy, and direct SQL (Subqueries, Stored Procedures, Triggers, Cursors) to manage relational databases including MySQL, PostgreSQL, and Oracle.
- Implemented Azure Storage solutions for scalable and secure data storage.
- Developed batch processors and Python scripts for ETL tasks, web scraping, automated data updates, and backend processing workflows.
- Worked extensively with JSON, XML and data-parsing routines for secure, structured data interchange across systems.
- Built data analytics and visualization components using Pandas, NumPy, Matplotlib, and HighCharts for exploratory data analysis and reporting.
- Implemented distributed data processing using PySpark, data frames, and SQL queries for faster computation on large datasets.
- Designed interactive and responsive UI components using HTML5, CSS3, Bootstrap, JavaScript, jQuery, AJAX, and contributed to front-end modules built with React.js, Vue.js and Angular.js.
- Utilized AWS services, including EC2, RDS, S3, and Lambda, for scalable and reliable cloud-based solutions.
- Integrated RabbitMQ messaging queues for internal workflows and alert systems, facilitating reliable asynchronous communication between microservices.
- Used JIRA for agile sprint tracking and Confluence for maintaining comprehensive project documentation, improving transparency and collaboration.
- Actively participated in Agile Scrum/Kanban ceremonies, sprint planning, daily stand-ups, and progress reporting using tools like JIRA and Confluence.

Environment: Python, Django, Flask, MySQL, PostgreSQL, Oracle, Azure, JSON, XML, Pandas, NumPy, Matplotlib, HighCharts, PySpark, HTML5, CSS3, Bootstrap, JavaScript, jQuery, AJAX, AWS, RabbitMQ, JIRA

Charter Bank, India
Full Stack Developer

February 2021 - March 2022

- Analyzed business requirements, created use cases, and prepared both low-level and high-level design documents.
- Developed and maintained web applications and portals using Python and Django, implementing dynamic content management features.
- Designed and implemented data management systems using MySQL, including database schema design and optimization.
- Built interactive dashboards and data visualizations using Matplotlib, HighCharts, Numpy, and other Python libraries to support business decision-making.
- Developed Python APIs for processing and debugging complex data structures and integrating backend services.
- Created GUI/web interfaces for dynamically displaying Python code documentation and other application features via browser.

- Implemented responsive frontend interfaces using HTML5, CSS3, JavaScript, and jQuery following web standards and best practices.
- Wrote Python scripts for calling external APIs and performing backend data processing tasks.
- Worked on AWS cloud services for ETL pipelines and scalable application deployment, handling data from heterogeneous sources.
- Created Python scripts for REST API integration and developed RESTful API endpoints using Django Rest Framework with ViewSets, Routers, and regex-based routing.
- Parsed and processed JSON and XML data efficiently using Python.
- Wrote raw SQL queries in Python to support database-agnostic operations across MySQL and Oracle databases.
- Worked in an Agile development environment, handling multiple high-priority tasks, participating in sprint planning, and reviewing project deliverables.

Environment: Python, Django, SQL, MySQL, AWS, JSON, XML, HTML5, CSS3, JavaScript, jQuery, Matplotlib, HighCharts, NumPy, Oracle, Agile and Windows.

Lupin Pharmaceuticals Limited, India

Dec 2019 - Feb 2021

Python Developer

- Responsible for gathering requirements, system analysis, design, development, testing, and deployment of web-based applications.
- Implemented business logic, data exchange, XML processing, and graphics creation using Python and Django.
- Developed views and templates in Django, leveraging its MVC architecture to create user-friendly and responsive web interfaces.
- Built frontend components using HTML5, CSS3, JavaScript, AngularJS, jQuery, and JSON to enhance application usability and performance.
- Integrated AJAX and jQuery for seamless asynchronous data handling between frontend and backend systems.
- Worked extensively with standard Python libraries such as NumPy, SciPy, Matplotlib, Pandas, and PyTables for data analysis, statistical computations, and reporting.
- Utilized Pandas for data analysis and generating insights from clinical, pharmaceutical, and operational datasets.
- Applied AJAX and jQuery for JSON-based data communication between frontend and backend services.
- Designed and optimized SQL queries, stored procedures, and triggers using Oracle SQL and PL/SQL for efficient data management.
- Followed Agile (Scrum) methodology, participating in daily stand-up meetings and implementing Test-Driven Development (TDD) to ensure high-quality code delivery.

Environment: Python, Django, XML, MVC, HTML5, CSS3, JavaScript, jQuery, JSON, AJAX, Numpy, Pandas, Matplotlib, Scipy, PyTables, Oracle, SQL, PL/SQL, Agile and Windows.

Education:

- ❖ Masters in information technology management, **Campbellsville University, Louisville, Kentucky.**
- ❖ Bachelors in mechanical engineering, **Kakatiya Institute of Technology & Science, Warangal, India.**